HAO ZHANG

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Research Interests: Expressive Speech Translation, Full-duplex / Conversational Naturalness, Sentiment Analysis, Low Resource Language

EDUCATION

Johns Hopkins University, USA, MSE in Computer Science 01/2025-12/2026
- GPA: **3.88**/4
- Courses: Machine Learning for Signal Processing (A+), Audio Signal Process (Audit), Information Retrieval and Web Agents (A), Introduction to Human Language Technology (A+), Natural Language Processing (A), Information Extraction (A+)
- Human Language Technology Concentration (Thesis Based)

Shandong University, China, B.Eng in Intelligent Medical Engineering 09/2020-07/2024
- GPA: **88.16**/100; WES Calculator GPA: **3.74**/4; Jr./Sr. GPA: **92.06**/100 (**3.93**/4)
- Courses: Python Programming and Introduction to Algorithms (A), Signal Analysis and Processing (A), Digital Image Processing (A), Healthcare Big Data Analytics (A), Cognitive science and brain-like computing (A), Machine Learning (A)
- Honors Graduate Student in Shandong University (Top 5 in Major) 2024, Academic Scholarship in 2022, 2023

PAPER & PATENT

Journal & Conference (†indicates equal contribution; ‡indicates corresponding author)

Speech & Language Processing (M.S.E.)

[1] **Hao Zhang**, Rui Liu, Chengzhe Li, Philipp Koehn, Learning Sentiment Dynamics: Knowledge-Augmented Sequence Modeling for Document-Level Aspect-Based Sentiment Analysis, *The 2027 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2026.10 (**Under Review**)

[2] **Hao Zhang**, ..., Laureano Moro-Velázquez, CONTACT: A Human-Grounded Benchmark and Surprisal-Based Predictive Scorer for Conversational Naturalness, *The Fifteenth International Conference on Learning Representations (ICLR)*, 2026.9 (**Under Review**)

[3] **Hao Zhang**, Ismail Rasim Ulgen, Bismarck Bamfo Odoom, Rui Liu, Zihan Zhang, Berrak Sisman, Philipp Koehn, Holistic Parallel Supervision for Expressive Speech to Speech Translation, *2027 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.9 (**Under Review**), Demo and Codes

[4] Zihan Zhang, Shreeram Suresh Chandra, Zongyang Du, Xiutian Zhao, Aurosweta Mahapatra, **Hao Zhang**, Philipp Koehn, Berrak Sisman, TRACE-EVC: Text-Guided Relative Affective Control for Zero-Shot Emotional Voice Conversion, *IEEE Open Journal of Signal Processing*, 2026.9 (**Under Review**), arXiv:2607.03666

[5] Sathvik Napa Ugandhar†,‡, **Hao Zhang**†,‡, ..., Laureano Moro-Velázquez‡, TRACE: Temporal Relationship-Aware Conversational Entrainment Detection in Dyadic Speech, *IEEE Spoken Language Technology Workshop (SLT)*, 2026.8 (**Accepted**), arXiv:2606.30543

[6] **Hao Zhang**, Thomas Thebaud, Georgi Tinchev, Venkatesh Ravichandran, Laureano Moro-Velázquez, TurnNat: Automatic Evaluation of Turn-Taking Naturalness in Dyadic Spoken Dialogue, *arXiv preprint*, 2026.7, arXiv:2607.01345, Demo and Codes

[7] Thomas Thebaud, Yuzhe Wang, **Hao Zhang**, Sathvik Manikantan Napa Ugandhar, Ashish Hallur, Georgi Tinchev, Venkatesh Ravichandran, Laureano Moro-Velázquez, SPEARBench: A Benchmark for Naturalness Evaluation in Streaming Speech-to-Speech Language Models, *Computer Speech & Language*, 2026.7, (**Under Review**) arXiv:2607.05365

Signal Processing & Time Series Analysis (B.Eng.)

[8] **Hao Zhang**, Teng. Li‡, Ai Jing, Siyuan Yang, Sequence-Spectrogram Fusion Network for Wind Turbine Diagnosis through Few-Shot Time Series Classification, *Advanced Engineering Informatics (AEI)*, (Impact Factor: 11.5), 2025.1 (**Published**), Codes and Datasets

[9] **Hao Zhang**, Zhendong Pang, Jiangpeng Wang, Teng Li‡, Few-Shot Learning Using Data Augmentation and Time-Frequency Transformation for Time Series Classification, *5th International Conference on Robotics, Intelligent Control and Artificial Intelligence (RICAI)*, IEEE, 2023.12 (**Published, Oral**)

[10] Zhendong Pang, **Hao Zhang**, Teng Li‡, Hybrid Fine-Tuning in Large Language Model Learning for Machinery Fault Diagnosis, *22nd IEEE International Conference on Industrial Informatics (INDIN)*, 2024.6 (**Published**)

[11] Bingyu Yang, ..., **Hao Zhang**, Teng Li‡, Hierarchical Informative Path Planning for Active Sensing and Field Reconstruction, *Proceedings of the 2024 3rd International Symposium on Control Engineering and Robotics (ISCER)*, ACM, 2024.8 (**Published**)

Invention Patent A Method and System for Wind Turbine Condition Monitoring Based on Multimodal Feature Fusion, 2024.8 (**Published**)

ACADEMIC RESEARCH

***Research Assistant,* Human Language Technology Center of Excellence (HLTCOE)** 08/2026 - Present
Supervisor: Prof. Kevin Duh, Berrak Sisman, Philipp Koehn, Johns Hopkins University

❖ **Enhancing LLMs to Translate Low-Resource Languages** 08/2026 - Present
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***Research Assistant,* Center for Language and Speech Processing (CLSP)** 11/2025 - Present
Supervisor: Prof. Laureano Moro-Velázquez, Johns Hopkins University

❖ **CONTACT: A Human-Grounded Benchmark and Surprisal-Based Predictive Scorer for Conversational Naturalness**
Funded by Amazon 11/2025-09/2026
- Led the design and early deployment of a naturalistic huma-AI speech interaction pipeline, integrating human-subject workflow, participant recruitment, recording infrastructure, pilot validation, and evaluation design.
- Designed controlled topic and assignment protocols and evaluated SoTA real-time speechLLMs (e.g. gpt-realtime, Nova 2 Sonic, and cascaded baselines) while minimizing confounds from topic repetition, ordering effects, and speaker-specific variability.
- Developed structured human-evaluation rubrics to assess conversational naturalness, paralinguistic cue understanding, emotion entrainment, and other fine-grained interaction quality.
- Designed a scalable platform concept for automated real-time human-AI benchmarking and built a reliable single-device recording pipeline with VB-Audio and Zencastr; Conducted a pilot study with 7 participants averaging 70 minutes per session.

❖ **TurnNat: Automatic Evaluation of Turn-Taking Naturalness in Dyadic Spoken Dialogue** 05/2025-07/2026

❖ **TRACE: Temporal Relationship-Aware Conversational Entrainment Detection in Dyadic Speech** 12/2025-03/2026

- Built DyadEE, a dyadic speech benchmark for emotional entrainment detection, totaling 21,596 dyads samples (1,143.81 h).
- Designed and implemented artifact-robust augmentation pipelines for entrained dyads via voice conversion (VC) and speech denoising, producing high-quality VC samples (105.58 h) and denoised samples (63.82 h) to encourage models to learn interaction dynamics rather than synthetic shortcuts.
- Contributed to the controlled construction of non-entrained dyads, including partner swapping and emotion-contradicting resynthesis, as well as dataset quality control and balance over relationship and conversational context using stratified sampling.

Supervisor: Prof. Philipp Koehn & Prof. Berrak Sisman, Johns Hopkins University

❖ **Holistic Parallel Supervision for Expressive Speech to Speech Translation** 03/2026-09/2026

- Identified that existing S2ST systems suffering from data that never jointly parallel in
- Designed VC-Dub: professional dubbing for human-performed content, prosody and timing, plus multilingual voice conversion for speaker matching, with real human speech kept as the target (speaker similarity 0.06 to 0.41; 210k pairs En-De/En-Es, 434 h).
- Built DirectS2ST (1.2B, end-to-end codec prediction from source speech only). VC-Dub improved all three S2ST model families, and DirectS2ST gained the most (expressiveness MOS 2.13 to 3.42, best prosody and pause scores.

Supervisor: Prof. Philipp Koehn, Johns Hopkins University

❖ **Learning Sentiment Dynamics: Knowledge-Augmented Sequence Modeling for Document-Level Aspect-Based Sentiment Analysis (DL-ABSA)** 05/2025 - Present

- Designed an ABSA framework integrating FLAN-T5 to improve document-level aspect-based sentiment classification (ASC).
- Built domain-specific knowledge bases via Chain-of-Thought (CoT) prompting to generate aspect candidates with GPT-5.
- Introduced the concept of aspect triggers, enabling less ambiguous and more fine-grained identification of aspect occurrences across long documents.
- Devised Next Polarity Prediction to explicitly modeling sentimental intensity sequence among reviews using the decoder with teacher forcing supervised by knowledge base matched pseudo label, addressed cross-sentence sentimental conflicts.

***Research Assistant,* Key Laboratory of Machine Intelligence & System Control, Ministry of Education** 07/2023-06/2024

Supervisor: Prof. Teng Li, Shandong University

❖ **Sequence-Spectrogram Fusion Network for Machinery Diagnosis through Few-Shot Time-Series Classification** 09/2023-06/2024

- Proposed a novel network consisting of scalable blocks for hierarchical multimodal fusion and joint representation learning.
- Devised an embedded hybrid encoder combining multi-head self-attention (MSA) and convolution module to enhance feature extraction, opti- mize the alignment of latent spaces, and allocate modalities weight adaptively for quality feature fusion.
- Conducted thorough comparative and ablation studies, demonstrating the outperformance and interpretability of our network.
- Yielded 0.996 macro AUC on 2 multi-class few-shot wind turbine fault datasets.

❖ **Few-shot Learning using Data Augmentation and Time-Frequency Transformation for Time Series Classification** 07/2023-09/2023

- Organized lab members to research overfitting issues of deep learning and few-shot learning (FSL) techniques.
- Proposed a data augmentation method while preserving temporal dynamics: converting time-series data into time-frequency (T-F) representations using short-time Fourier transform (STFT), and applying random erasure on T-F images to increase data.
- Proposed a feature-level multimodal network aligning time-series and images to exploit and merge complementary features.
- Demonstrated our proposed FSL framework on medical and industrial few-shot datasets with over 95% F1 scores.

WORK EXPERIENCE

***Algorithm Engineer Intern,* iFLYTEK CO.LTD.** 07/2024-11/2024

Business Unit: Xunfei Healthcare, Research Institute

❖ ***Xiaoyi* Medical Interrogation LLM Optimization: Data-Centric LLM Fine-Tuning for Patient Persona Slot Extraction**

- Designed Chain-of-Thought prompting strategies to integrate GPT-4o into the data pipeline, including: (1) generating synthetic dialogues to compensate for missing labels or under-represented slot data, and (2) performing efficient quality screening, correction, and standardization of dialogues pre-annotated by the deployed *Xiaoyi* LLM.
- Developed a human-in-the-loop annotation framework, streamlining collaboration with doctors and crowdsource annotators through automated task routing and data recollection. Reduced rework ratio by 14% and saved 110 person-hours per month.
- Constructed a high-quality supervised fine-tuning corpus from 10GB+ online dialogues, applied instruction-tuning to post-train *Xiaoyi*, and achieved 10%+ F1 improvement in average across 7 critical patient slots.

TEACHING EXPERIENCE

- Course Assistant, Johns Hopkins University, 2026 Fall, Natural Language Processing (Supervisor: Jason Eisner)
- Teaching Assistant, Shandong University, 2023 Spring, Machine Learning (Supervisor: Teng Li)

SKILLS

Computer Language: Python/Matlab (advanced), C/C++/SQL/R/Java (intermediate).
Framework, Package, and Tool: Pytorch, Pytorch Lightning, HuggingFace, transformers, PyWavelets, SciPy, NLTK, OpenCV, sktime, Matplotlib, WFDB, Git.

EXTRACURRICULAR HIGHLIGHTS

Student Admission Reviewer (MSE of Computer Science), Department of Computer Science, Johns Hopkins University
Academic Reviewer

- Journal: Knowledge-based Systems, MethodsX
- Conference: 2024 International Conference on Biomimetic Intelligence and Robotics & Medical Robot Forum (ICBIR).

Oral Presentation at 5th International Conference on Robotics, Intelligent Control and Artificial Intelligence (RICAI).